

Predictive Analytics Using Historical Data

Key Insights:

1. Historical Data Shows Clear Trends

Exploratory Data Analysis revealed a measurable trend in sales behavior, making the dataset suitable for predictive modeling techniques.

2. Regression Successfully Captures Relationships

The Linear Regression model identified relationships between time-based features (Year, Month, Day) and sales performance, enabling future value prediction.

3. Low Prediction Error

Evaluation metrics such as MAE and RMSE indicate that the model maintains relatively low prediction errors, demonstrating acceptable forecasting accuracy.

4. High Correlation Between Actual and Predicted Values

The Actual vs Predicted scatter plot shows data points clustered around the ideal prediction line, confirming that the model generalizes well to unseen data.

5. Forecasting Reveals Future Business Trends

The future forecast visualization indicates expected sales patterns for upcoming months, helping stakeholders anticipate future performance.

6. Importance of Data Preprocessing

Data cleaning, duplicate removal, and date feature extraction significantly improved model quality and prediction reliability, highlighting the importance of preprocessing in machine learning workflows.